

Ronaldo Franco Jaldin

Machine Learning Research Engineer

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Research Profile

Machine learning research engineer focused on LLMs for software engineering, retrieval, and GFlowNets.

Education

BSc, Applied Mathematics and Informatics — HSE University

Moscow · Sep 2021–2026

Thesis (2025–26): *GFlowNet Post-training of a Language Model for Text-conditioned Molecule Generation*

Honors

- ICPC World Finals — Finalist (2022); Topcoder Open 2019 (TCO-19), South America Regional — 2nd Place.
- ICPC South America Regional — 3rd (2020), 10th (2019), and 15th (2018); Bolivia Preliminary Contest — 1st Place (2018, 2019, 2020, and 2025).
- International Olympiad in Informatics (IOI) — participant (2014, 2015, and 2017).
- Ibero-American Olympiad in Informatics — Bronze (2014 and 2015); Bolivian Olympiad in Informatics — Gold (2016) and Silver (2013, 2014, and 2015).

Publication

Exploring the Potential of Conversational Test Suite Based Program Repair on SWE-bench

A. Cheshkov, P. Zadorozhny, R. Levichev, E. Maslov, and **R. Franco Jaldin**

arXiv:2410.04485 · 2024

Evaluated test-feedback-driven repair on 92 selected SWE-bench Lite problems; with oracle function localization, Llama 3.1 patches passed public and hidden tests for 47 % of problems versus 34 % for repeated generation at equal request budget.

Research Experience

Research Assistant — Huawei Research Institute

Moscow, Russia · Mar 2024–Mar 2025

- Built a CodeXGLUE-based embedding pipeline for patch and file retrieval; fine-tuned Jina embeddings with QLoRA and outperformed the off-the-shelf baseline, reaching 0.62 MRR on SWE-bench Verified.
- Optimized prompts and multi-trial sampling for Llama 3.1 issue resolution, obtaining 77 % resolution when the model was supplied with oracle-retrieved context.
- Developed agentic and agentless pipelines for constructing reproducible test environments across repositories; contributed Docker environments to the SWE-bench harness.
- Mined and cleaned issue–patch–file pairs from open-source repositories to build supervised training data for code-retrieval models.

Student Researcher — HSE AI for Computational Biology Lab

Moscow · Mar–Sep 2023

- Worked in C++ on parallelization and concurrency optimizations for Crux, an open-source toolkit for peptide identification from tandem mass-spectrometry data.
- Studied foundational computational-proteomics models, including the GLEAMS Siamese network, contrastive learning, and learned mass-spectrum embeddings for large-scale clustering.

Research Projects

GFlowNet post-training of BioT5+

BSc thesis · 2025–26

- Post-trained BioT5+ as a GFlowNet using LoRA, a scalar flow head, and Detailed Balance to sample diverse drug-like molecules conditioned on natural-language descriptions.
- Compared on-policy, reward-prioritized, and TB-residual-prioritized replay strategies; combining supervised reference trajectories with on-policy rollouts produced the most stable training behavior.

Identity recognition from eye movements

HSE University · Nov 2022–Jul 2023

- Compared participant classification and triplet-based embeddings on GazeBase; adapted convolutional, recurrent, and residual models to gaze sequences, with supporting MNIST retrieval experiments.

Natural-language code search with UniXcoder

Python · PyTorch · Transformers

- Adapted retrieval experiments for Java and Python; explored function/window embeddings, maximum-similarity aggregation, and length-based ranking analysis.

Applied Machine Learning Experience

Freelance Machine Learning Engineer

Bolivia / Remote · Apr 2025–present

- Developing a computer-vision system for a Bolivian coffee shop that analyzes security-camera video to classify prepared items, count production, verify customer hand-offs, and generate time-stamped sales reports. Building the domain dataset from scratch using model-assisted annotation.
- Building, since Sep 2026, a WhatsApp interface for Odoo Community Edition that answers natural-language questions about inventory quantities and stock movements; responsible for test design, evaluation, and measurement of NLP response quality.
- Delivered a production face-attendance system for a private company using standard cameras, combining face detection and anti-spoofing checks designed to prevent fraudulent check-ins.

Teaching, Mentorship and Community Service

Scientific Committee — International Olympiad in Informatics (IOI) 2025

Bolivia · 2025

Problem setting, testing, and contest design for the international competition.

Course / activity	Institution / format	Classes	Year
Introduction to Machine Learning I	UMSS, Cochabamba	8	2026
Introduction to LLMs	Private classes, online	10	2025
Advanced Competitive Programming Workshop	UMSS, Cochabamba	4	2025
ICPC Training Camps (July / December)	UMSS, Cochabamba	16	2018–19
High-school coaching and ICPC co-coaching	Various, Bolivia	20+	2016–23

- Machine learning:** neural networks and backpropagation; MLP from scratch on MNIST; matrix calculus, gradients, and Jacobians; optimization; dropout and batch normalization; cross-validation and model selection.
- LLMs and deep learning:** transformers; pre-training and tokenization; scaling and limitations; supervised fine-tuning with LoRA; alignment; LLM agents and retrieval-augmented generation.
- Algorithms and competitive programming:** graph algorithms; data structures; dynamic programming and its optimizations; number theory; probability; geometry; strings; FFT/NTT and generating functions.
- Organizer and problem setter:** SanSi Cup Bolivia and the 2023/2024 individual national programming contests at UMSS, with more than 200 participants, supporting Bolivia’s competitive-programming community.

Technical Skills

Languages	Python · C++ · SQL · Bash
ML / LLM / RL	PyTorch · Hugging Face Transformers · PEFT · vLLM · Unsloth · LoRA/QLoRA · Gymnasium · Pandas
Computer vision	OpenCV · YOLO · ByteTrack · ONNX Runtime
Retrieval & agents	sentence-transformers · FAISS · LlamaIndex · LangGraph
Methods	LLM post-training · policy optimization · GFlowNets & amortized inference · embedding retrieval · SWE agents
Systems	C++ concurrency · multithreading · performance optimization
Evaluation	scikit-learn · Weights & Biases
Backend / DevOps	Docker · FastAPI · Pydantic · PostgreSQL · Git/GitHub · GitHub Actions
Integrations	Odoo Community Edition · WhatsApp Cloud API

Languages

Spanish (native) · English (fluent) · Russian (intermediate). Teaching experience in Spanish and English.